

Design of Enhanced AMHS with Color Sorting Integration Using 3D Printing Technology

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Abstract:

Material handling is a significant challenge in industries, often requiring substantial manpower and labor. To address this, we propose a color sorting mechanism that leverages advanced sensors and robotics to classify and separate materials based on color. This technology is extensively used in industries such as recycling, agriculture, food processing, and manufacturing, where precise sorting enhances quality control and reduces waste. By minimizing human intervention, the system improves accuracy, reduces errors, and optimizes production efficiency. Additionally, the integration of conveyor-based or robotic sorting mechanisms enhances automation, making industrial operations more sustainable and cost-effective.

This prototype is designed using Fusion 360 and 3D printing for structural components, while the electronics are controlled by an Arduino with three degrees of freedom achieved through servo motors, an electromagnetic end effector. Incorporating Smart Warehousing and Industry 4.0 principles, the system enhances automation in material handling. The color-sorting mechanism utilizes advanced sensors to detect subtle color variations, enabling high-speed and precise material classification. This approach optimizes efficiency, reduces human intervention, and improves accuracy in industrial applications such as recycling, manufacturing, and logistics.

Keywords: SolidWorks, TCS 34725 Sensor, Articulated robot, Servo motor

Introduction:

The field of robotics is a branch of engineering and computer science that deals with the design, construction, and operation of robots. It is an interdisciplinary field that draws upon knowledge and techniques from various other fields, including mechanical engineering, electrical engineering, computer science, and artificial intelligence. The goal of robotics is to create machines that can perform tasks autonomously, with minimal human intervention. This includes tasks that are too dangerous, too difficult, or too repetitive for humans to perform. Robotics has already had a significant impact in areas such as manufacturing, space exploration, and healthcare, and it is expected to have an even greater impact in the future. One of the key challenges in robotics is developing machines that can interact with their environment and adapt to changing conditions. This requires the use of sensors and other technologies that allow robots to perceive and interpret their surroundings. Robotics also

involves the development of algorithms and software that allow robots to make decisions and carry out tasks autonomously.

Conventional grippers struggle to efficiently sort metallic waste due to variations in shape, size, and weight, leading to slower processing, increased error rates, and potential damage to robotic arms and surrounding equipment. Their limited flexibility restricts handling diverse objects, reducing throughput and overall efficiency, while inadequate gripping poses safety risks by causing dropped objects that endanger personnel and machinery. Additionally, the TCS3200 color sensor has a narrow detection range and is sensitive to ambient light, making it prone to misclassification errors in metallic waste sorting. Unlike advanced sensors like the TCS34725, it struggles with subtle color variations and fluctuating lighting conditions, leading to inconsistent readings that compromise sorting accuracy and efficiency. These limitations collectively reduce the reliability of automated sorting systems and impact the quality of material segregation.

Researchers have explored various **color-based sorting mechanisms** to enhance industrial automation. **Lim Jie Shen** used a **TCS3200D color sensor**, **Arduino UNO**, and **servo motor** for conveyor-based sorting. **Snehal Shirgave** developed a **Zigbee-based system** with **Raspberry Pi** and WiFi for automation. **Rudresh H.G.** implemented a **robotic arm** to pick and place objects based on **color sensor data**, improving accuracy and reducing labor.

Advanced **sensor integration and IoT automation** further enhance sorting efficiency. **Shweta** used a **TCS34725 sensor** with a **microcontroller and motor driver** for sorting fruits and grains. **D A Jakkan** applied **IoT** for automation, integrating **power supply and regulation units**. **Himanshu Patel** developed a **food sorting system** to detect **rotten produce**, benefiting both **small- and large-scale industries**.

Low-cost automation solutions improve **sorting accuracy and efficiency**. **Ch. Shravani** designed an **affordable sorting machine** for small industries, reducing **manual labor**. **Uzma Amin** implemented a **PIC microcontroller (18F452)** with **SPI, I2C, and UART modules** for better control. These innovations collectively advance **industrial automation and material sorting**.

Conventional color sorting systems face challenges in accurately classifying and separating materials due to limitations in color detection and gripping mechanisms. Sensors with a narrow detection range struggle to differentiate subtle color variations, leading to misclassification and reduced sorting accuracy. Additionally, inefficient gripping systems contribute to errors, slow processing, and increased operational costs. These shortcomings hinder automation in industries such as recycling, manufacturing, and food processing, highlighting the need for an advanced, precise, and reliable color sorting system to enhance efficiency and reduce human intervention.

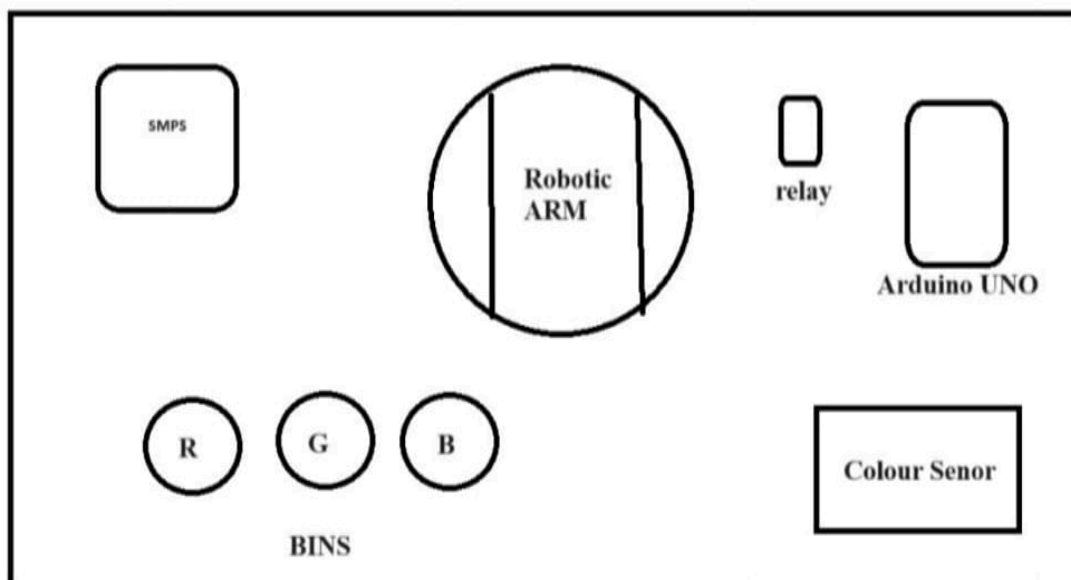
Conceptual Design:

The main aim of the project is to sort the objects by means of their colour. The sorting system is capable of recognising three colours Blue, red, green. The Triangular robot which is mounted on a platform is setup in the home position and picks the object at that position by colour recognition. The picking of the object is done by a magnetic gripper and is in 'ON' position until the object is sorted with respect to its color. Each colour sorting box is at a certain distance from the home position and after successful recognition of colour the distance from the home position is calculated and the robot is made to move to that certain distance. After successful sorting the robot is set to come back to its home position for the next operation.

Due to the mounting on the platform the no. of degrees of freedom of the robot is reduced by 1. Servo motors are used to move the height and distance arms according to the position.

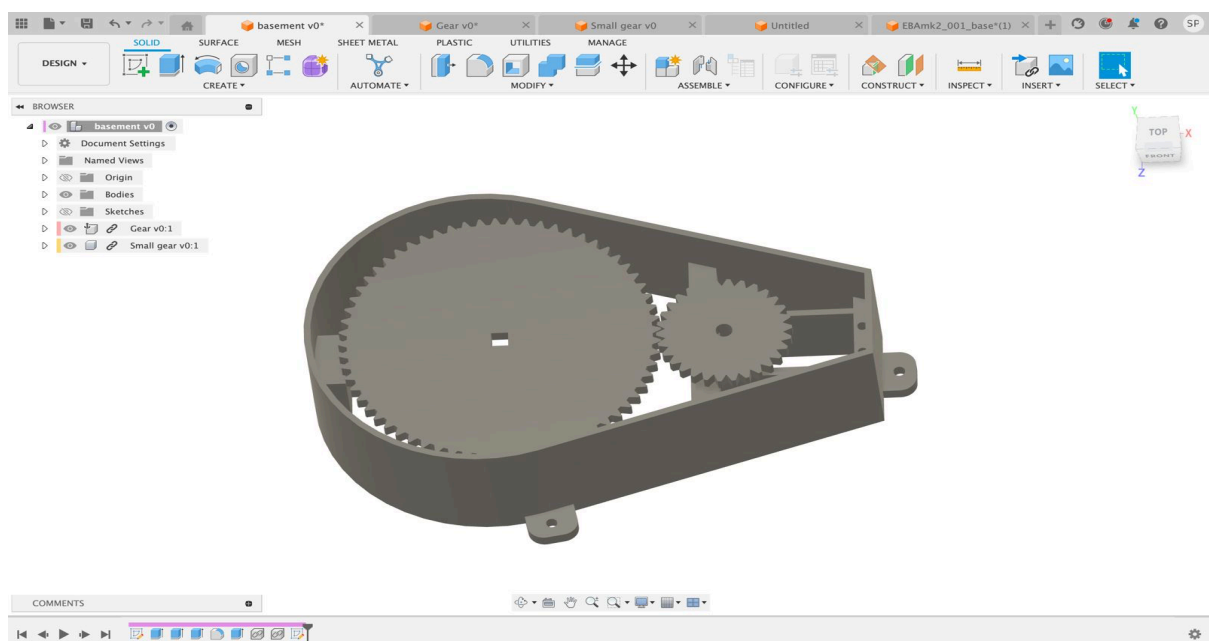
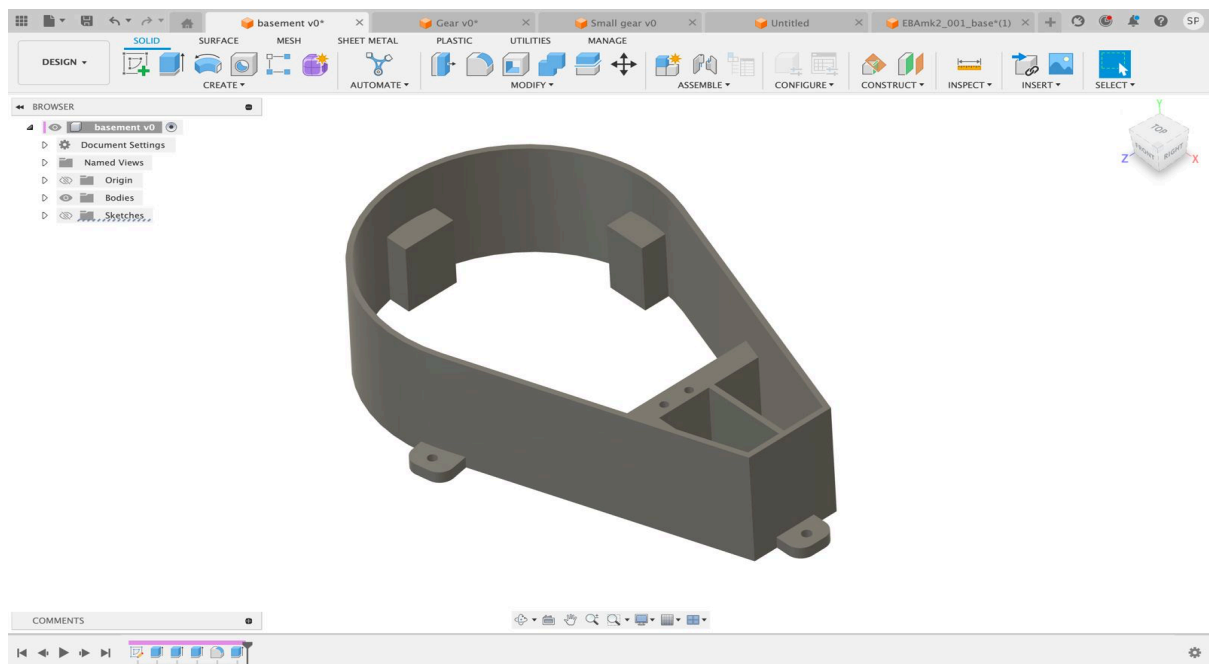
A TCS 34725 colour sensor is placed at the start and recognizes the colour of the component and with the help of serial communication, the colour of the component is communicated to the stepper motor controller to move to the sorting box associated with the colour recognised.

The order of sorting of colours in this system are decided to be Red, White and green. The distances from the home position are calculated and coded in the Arduino. For initializing the first operation we decided on a home position that is 25mm from the left end which is the position of the colour sensor and each time the robot does an operation it comes back to this position.



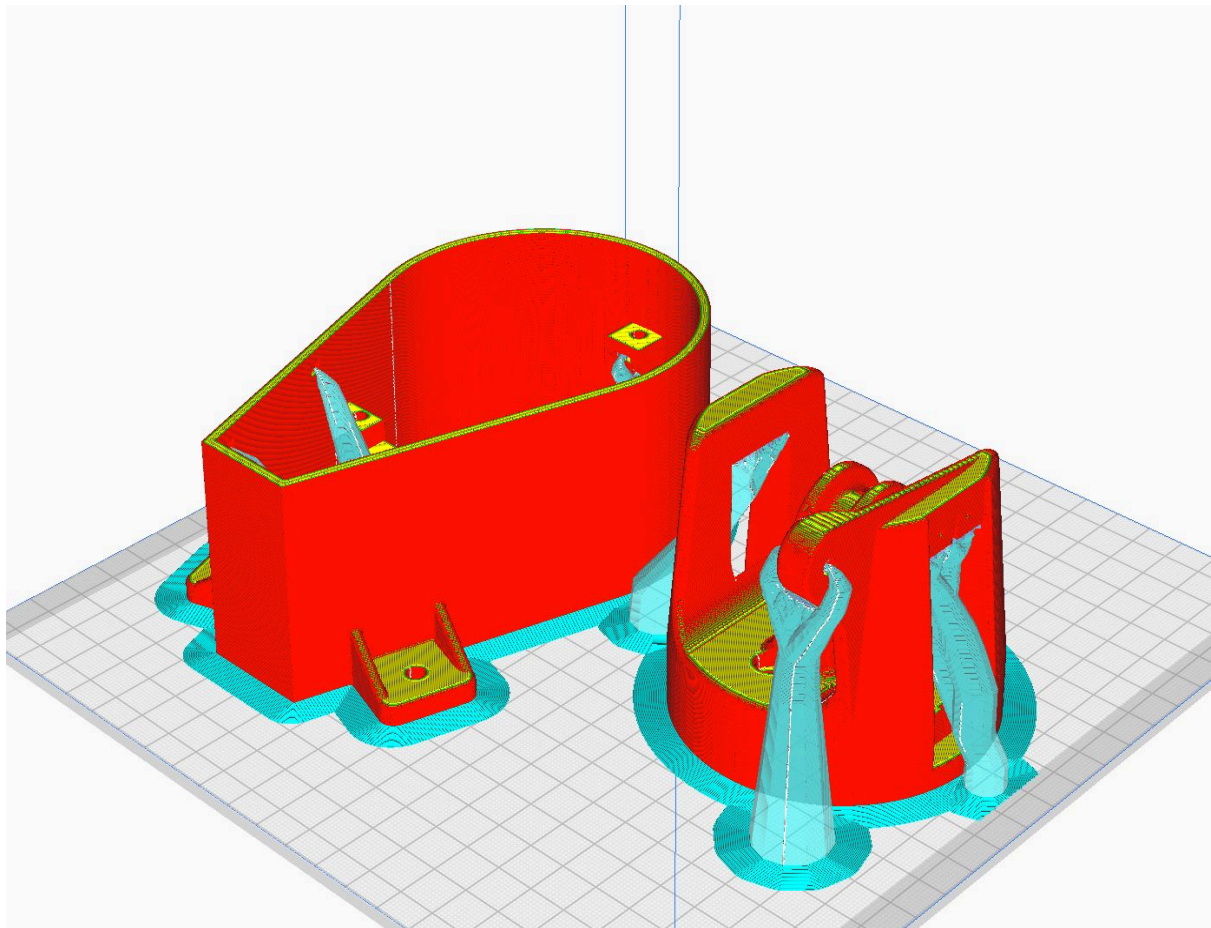
Modelling:

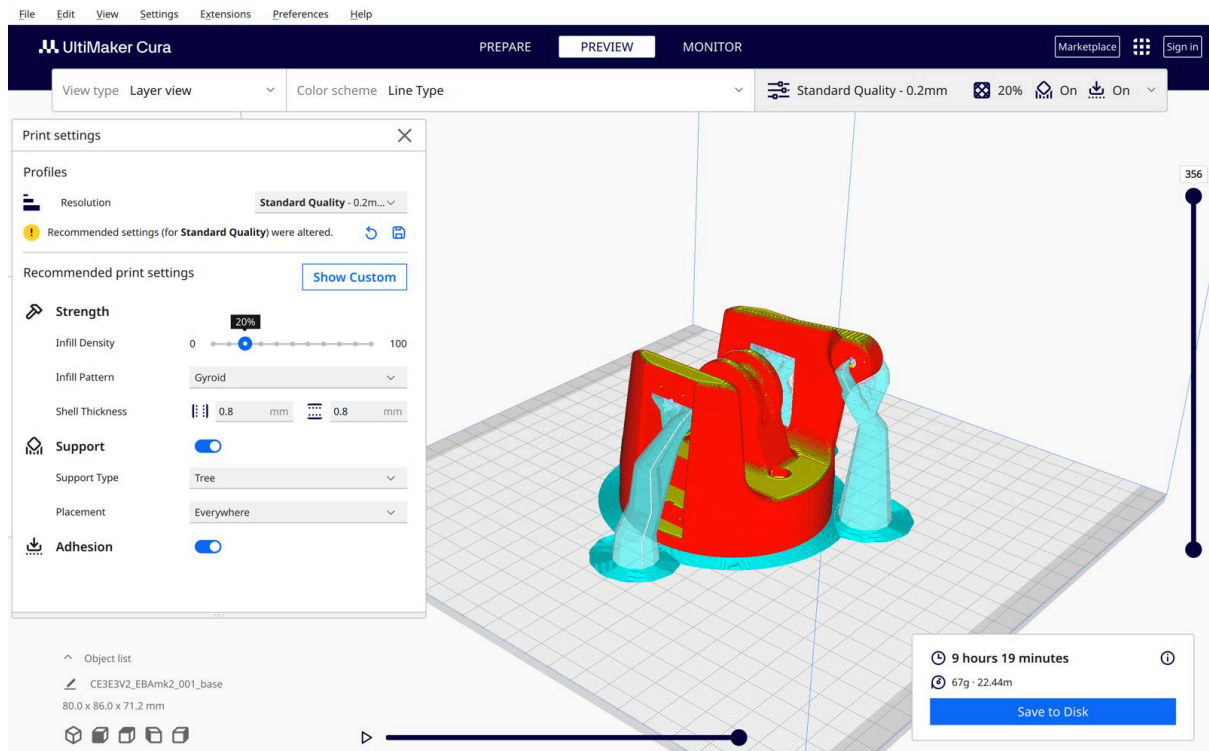
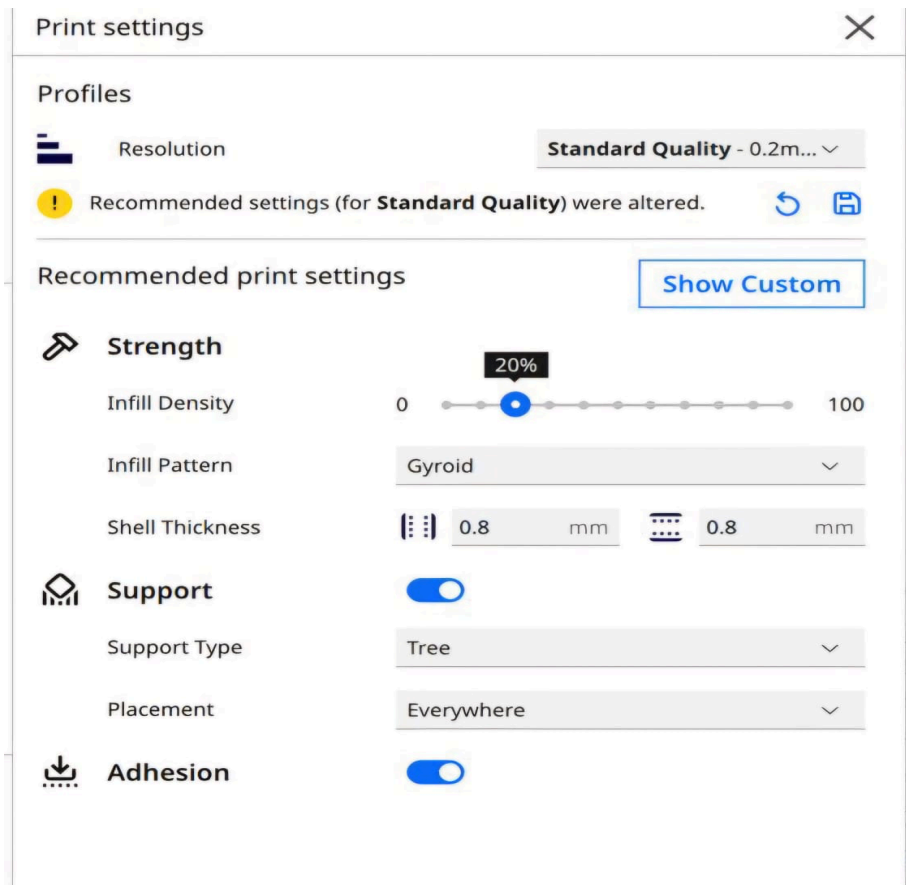
The modeling of the system is carried out using Fusion 360, enabling precise design and simulation of the components. This software facilitates the creation of detailed 3D models, ensuring accurate structural design and component integration. Fusion 360 also allows for design validation and modifications before fabrication, optimizing the overall efficiency of the system development process. Once a 3D model is created, it can be textured, lit, and rendered to create a realistic final image or animation. The resulting 3D models can be used in a variety of ways, including as assets for video games, animations for films and TV shows, and visualizations for architectural and engineering projects.



3D Printing:

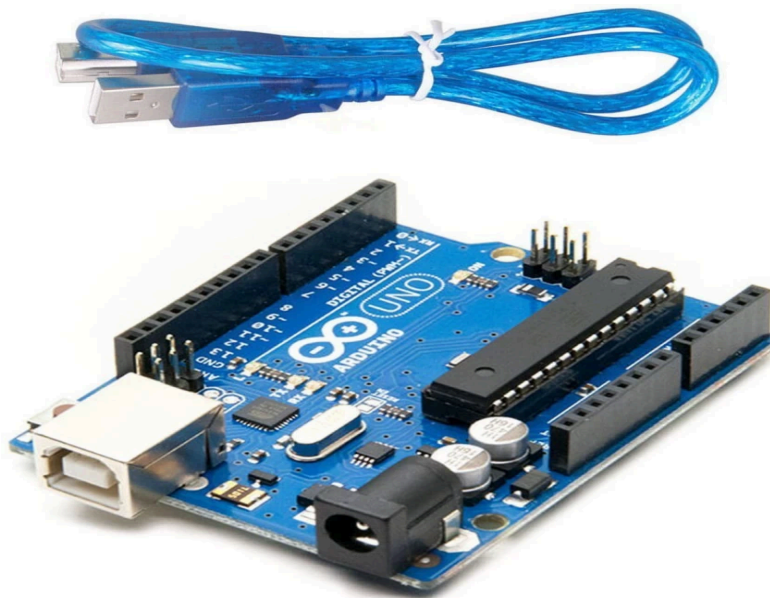
The digital model is first created using computer-aided design (CAD) software, ensuring precise dimensions, structural integrity, and design optimization. This model is then converted into a format compatible with 3D printers, which interpret the file and fabricate the physical object layer by layer using materials such as plastic. The process employs Fused Deposition Modelling (FDM), where a heated nozzle extrudes melted filament in successive layers, gradually building the object with high accuracy. This additive manufacturing technique allows for rapid prototyping, cost-effective production, and design flexibility, making it ideal for creating customized components. Additionally, 3D printing reduces material waste compared to traditional subtractive manufacturing methods, enhancing sustainability. The finalized 3D-printed structure serves as a robust prototype for further testing and refinement before full-scale implementation.



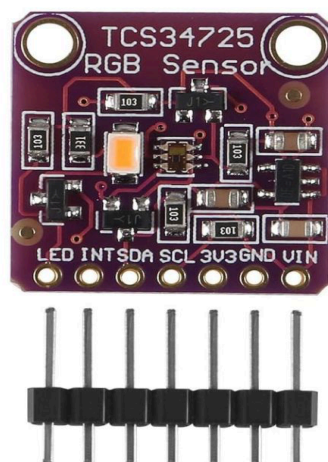


Electronics:

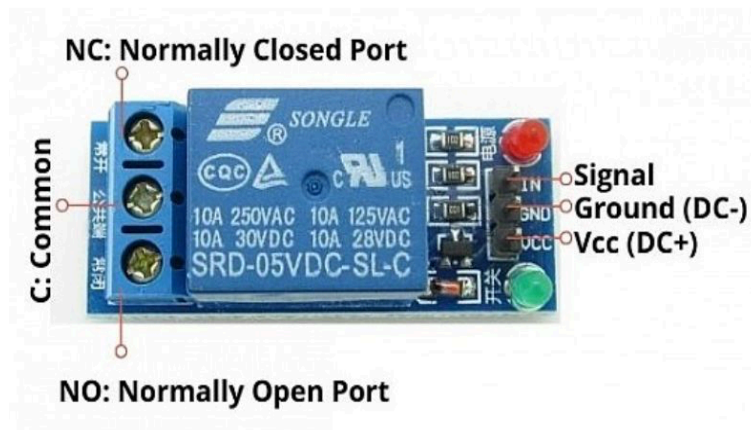
Arduino Uno: The Arduino Uno is a popular open-source microcontroller board featuring 14 digital input/output pins, 6 analog inputs, a 16 MHz quartz crystal, a USB connection, a power jack, and a reset button.



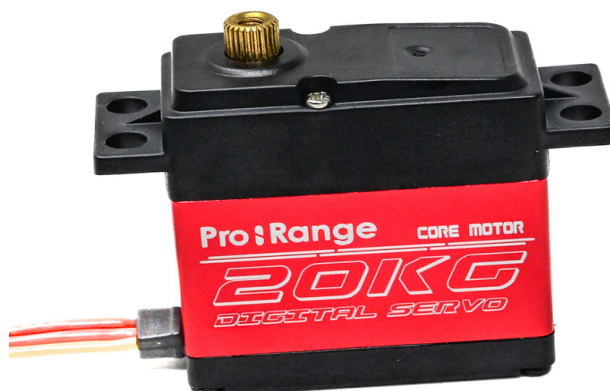
TCS 34725 colour sensor: The TCS34725 colour sensor enables detect a wider spectrum of colours with greater precision, making it well-suited for applications requiring accurate colour identification.



5V 1 CHANNEL MAGNETIC RELAY: A 5V 1 channel magnetic relay is an electronic component used to control high- power devices with a low-power signal. The relay consists of a coil that is energized by a small voltage (5V in this case) to create a magnetic field that switches the state of the contacts. The contacts are used to control the flow of current to the device being controlled.



Servo Motor: A 20kg servo motor operates using a PWM signal, where pulse width controls the angle. It has a DC motor, gears, and a potentiometer for position feedback, ensuring precise movement. The control circuit adjusts the motor based on feedback to reach the desired position. It typically operates at 6V-7.4V with a current draw of 1A-3A. Common applications include robotics, RC vehicles, and industrial automation.



Pseudo Code:

```
#include <Wire.h>
#include "Adafruit_TCS34725.h"
#include <VarSpeedServo.h>

// Initialize sensor and servos
Adafruit_TCS34725 tcs = Adafruit_TCS34725();
VarSpeedServo servo1, servo2, servo3;
int grip = 12; // Gripper pin

void setup() {
  Serial.begin(9600);
  servo1.attach(9);
  servo2.attach(10);
  servo3.attach(11);
  pinMode(grip, OUTPUT);

  home(); // Set to home position

  if (!tcs.begin()) {
    Serial.println("Sensor not found!");
    while (1);
  }
}

void loop() {
  uint16_t r, g, b, c;
  tcs.getRawData(&r, &g, &b, &c);
  delay(100);

  if (c >= 20) { // Object detected
    hold(); // Pick object

    if (r > b && r > g) moveTo("red");
    else if (b > r && b > g) moveTo("blue");
    else if (g > b && g > r) moveTo("green");
```

```

        drop(); // Release object
        home(); // Return to start
    }
}

void home() { setServo(90, 90, 90); }
void hold() { setServo(0, 120, 72); digitalWrite(grip, HIGH); delay(500); }
void drop() { setServo(servo1.read(), 110, 72); digitalWrite(grip, LOW); delay(500); }

void moveTo(String color) {
    if (color == "red") setServo(82, 106, 94);
    if (color == "green") setServo(126, 106, 94);
    if (color == "blue") setServo(162, 106, 94);
}

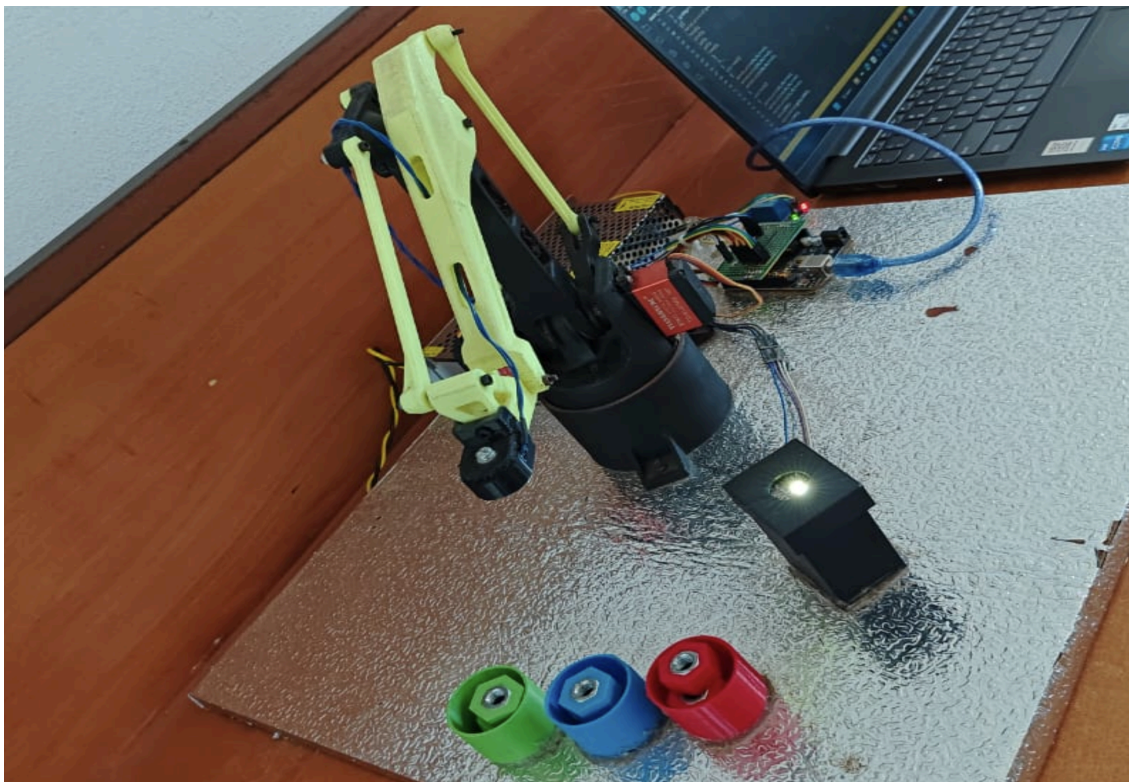
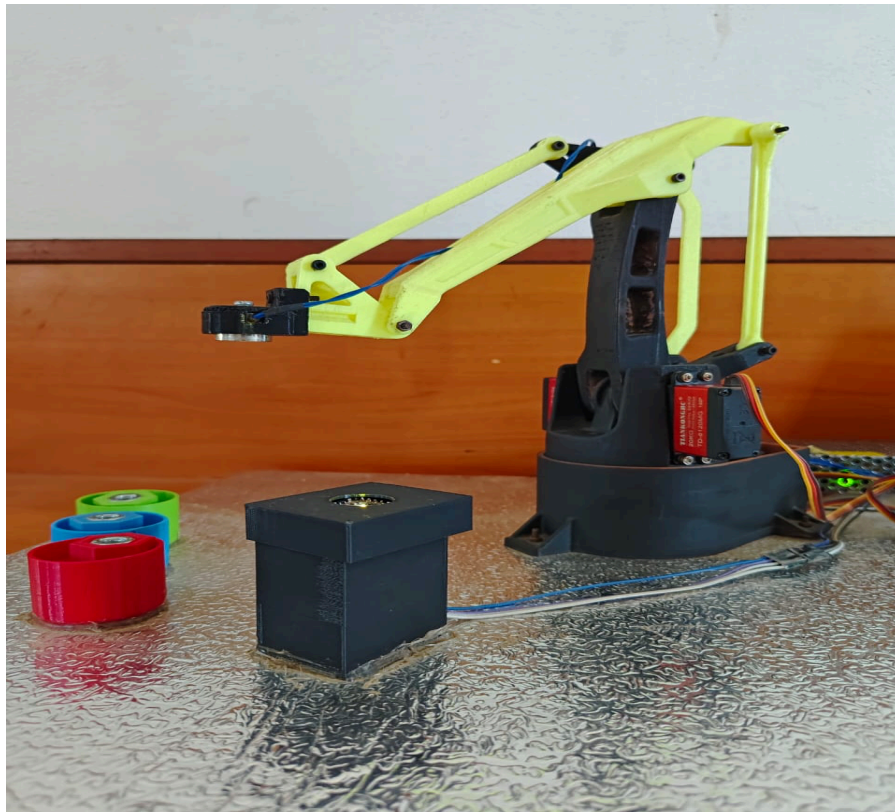
void setServo(int s1, int s2, int s3) {
    servo1.write(s1, 30, true);
    servo2.write(s2, 30, true);
    servo3.write(s3, 30, true);
    delay(500);
}

```

Results and Discussion:

The integration of the TCS34725 color sensor and magnetic gripper into the robotic arm system has significantly improved efficiency in sorting and handling metallic waste. The sensor's advanced color detection capabilities enable precise classification of metallic objects, addressing limitations of conventional sensors like the TCS3200, which struggle with subtle color variations. Additionally, its enhanced sensitivity to low-light conditions ensures consistent performance, optimizing sorting accuracy even in environments with limited illumination. The sensor's advanced color filtering further minimizes misclassifications, improving the overall reliability of material sorting.

Paired with the magnetic gripper, the robotic arm securely grasps metallic objects of different shapes, sizes, and weights, reducing the risk of dropped or misplaced items. The strong magnetic force enhances operational efficiency, minimizing manual intervention and streamlining the picking and placing process. By leveraging these advanced sensing and gripping technologies, the robotic arm system achieves higher accuracy and throughput, optimizing material handling and ensuring seamless operation.



Conclusion:

The color-based sorting system utilizing a robotic arm manipulator and a magnetic gripper presents a promising solution for automating sorting processes across various industries. The system has demonstrated its ability to efficiently and accurately sort objects based on their color. Throughout this journey, we encountered multiple challenges at different stages of execution. Our findings revealed that several factors could enhance the system's performance and accuracy, particularly through the incorporation of AI techniques. Additionally, the use of a parallel manipulator provides a high degree of flexibility and versatility in handling diverse objects and sorting criteria.

Overall, this color-based sorting system represents a significant advancement in the automation and optimization of sorting processes. While challenges remain, the potential benefits in terms of speed, accuracy, and productivity make it an effective solution for numerous industries. As technology continues to evolve, further advancements in robotics and automation are expected, potentially transforming the way we live and work.

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